Using Search Data to Crowd-source Unobserved Substitution Patterns for Demand Prediction

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Abstract

This paper develops methodology to enhance the accuracy of predictive demand models that rely on clickstream (or search) data, particularly in the context where only a limited set of product characteristics are observable. Traditional models often utilize the history of products browsed by consumers for constructing explanatory variables, and measure the relationship between these previously-browsed products and the target product through distance in a characteristic space. However, in many instances characteristics data are not available. To address this limitation, I propose a novel pairwise substitution measure derived from the aggregation of product searches across *all* consumers. This approach crowdsources the entirety of consumers' search behaviors to infer substitution patterns which aides in predicting consumer purchases. I estimate the model using a clickstream dataset from a large UK eCommerce platform, and show that incorporating this crowdsourced information on substitution patterns into a demand prediction model leads to effective predictions even without characteristics data. The models are estimated using both logistic regression and a state-of-the-art machine learning algorithm, LightGBM, and I show that the LightGBM estimation method performs better than logistic regression, and is able to take better advantage of the crowdsourced substitution patterns.

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Introduction

Consumers navigating online marketplaces typically search a variety of substitute goods before making purchases. The sequence of these searches, captured through clickstream data, provides a window into the decision-making process of consumers, revealing a "search set" of products considered before a purchase.[1] Many applications of clickstream data aim to predict consumers' product choices (j) as a function of their search sets (J) (Shiller (2020), Ruiz et al. (2020), Gabel and Timoshenko (2022)).

Effective predictive models necessitate that explanatory variables associated with each product in the search set ($j' \in J$) accurately reflect the influence of prior searches on the likelihood of purchasing a product (j). Specifically, accurate prediction requires these variable to reflect consumers' substitution patterns.[2] If demand-relevant characteristics are available, then the substitutability of j with each j' could be measured by the degree of similarity in their characteristics, following the characteristics-space approach. However, as pointed out by the marketing literature, there are significant costs associated with gathering detailed characteristic data alongside clickstream data for eCommerce platforms (Gabel and Timoshenko (2022)); for instance the popular com-Score web browsing data requires researchers to independently collect product characteristics (Bronnenberg et al. (2016), Shiller (2020)). Therefore such similarity metrics will in practice often be uninformative of *all* consumers' underlying substitution patterns.

To address this limitation, this paper develops a novel approach to crowdsource substitution patterns by leveraging search data from *all* consumers. I introduce a novel pairwise measure called the Joint Consideration (JC) measure, which aggregates the search sets of *all* consumers. The measure follows the rationale that if two products j and j' frequently appear together in many consumers' search sets, then they are likely close substitutes. Then the presence of product j' in a consumer's prior searches suggests a probable interest in j. This measure is straightforward to compute, and I show that using it as a measure of substitution leads to effective predictions of consumer choices.

The empirical analysis begins with a predictive demand model using characteristic-space measures of substitution, serving as a benchmark for cases with available demand-relevant characteristic data. Specifically, for each j' I observe three discrete characteristics and include indicators of identical characteristics with j as explanatory variables. Price differences between j and j' are also included. This benchmark model, hereafter called the characteristics model, relies on characteristic-space distances to gauge substitutability. Subsequently, I estimate a model that instead uses the JC measure between j and j' as the sole explanatory variable associated with each j'.

I estimate these models using a clickstream dataset from a UK e-commerce platform, that contains users' sequences of clicks - and purchases, if any - onto different product pages. Using these data I estimate demand-prediction models using logistic regression and LightGBM (Ke et al. (2017)), a cutting edge machine

[1] These products constitute a subset of the consumer's broader consideration set.

[2] In theory, j and j' could be either complements or substitutes, but in my application I find that consumers are more likely to search for substitute products, since most search sets result in the purchase of only one product. Search sets that result in multiple purchases (<1%) are dropped from the analysis.

learning method. Each model uses different measures of substitution as outlined above and their prediction performance, measured using common measures of fit in the machine learning literature, are compared. Notably, the LightGBM method significantly outperforms logistic regression when comparing models with the same explanatory variables. Moreover, LightGBM shows greater improvements in fit from using the JC measure. Together, these results confirm the findings of Bajari et al. (2015) who show that a closely related machine learning method, Random Forests, outperforms traditional econometric techniques such as logistic regression in demand prediction.

To assess each model's performance, I employ common fit metrics from the machine learning literature, such as the Area-Under-the-Curve (AUC) of the Receiver-Operating-Characteristic (ROC) plot, with values ranging from 0 to 1, where a model with no explanatory power would score 0.5. The JC model achieves an AUC of 0.8986, indicating strong predictive power. For comparison, Shiller (2020) also develops a predictive model using search data without using product characteristics to construct explanatory variable. They find an AUC of 0.71 when using a fixed-effects approach to measure the impact of past purchase behavior on purchase probability. The benchmark characteristics model achieves an AUC of 0.9053, illustrating that the JC model's performance is comparable. Additionally, a combined model using both distance in characteristics-space and the JC measure as explanatory variables (called the characteristics plus JC model) achieves an AUC of 0.9125. Comparing the performance of the JC model to Shiller (2020) suggests that when demand-relevant product characteristics are not available, using the JC measure to inform substitution is effective in building predictive demand models. Since the JC model performs comparably to the characteristics model, this suggests that the JC measure can be used to effectively crowdsource substitution patterns even in the absence of demand-relevant characteristics. Nonetheless, these approaches can be combined as in the characteristics plus JC model, so that when some demand-relevant characteristics are available incorporating the JC measure into prediction models can still lead to improvements in performance.

The relative performance of the models are similar when evaluated using misclassification rates, broken down by the misclassification rate of observed purchases - the False Negative Rate (FNR) - and the misclassification rate of observed non-purchases - the False Positive Rate (FPR). The JC model achieves 21.20% and 12.24% for each rate respectively, and the characteristics model achieves 20.70% and 12.05% for each rate respectively. Lastly, I find evidence that the JC measure's performance would scale well with more historical search data; when constructing the JC measure with a smaller sample, the AUC for the JC model is reduced to 0.8966.

To support the findings from the empirical analysis, I conduct a Monte Carlo simulation. In each period in the simulation setting, consumers are faced with the decision to either purchase a product in the product space or an outside option. The DGP is designed to mimic a consumer browsing and purchasing products in an online market, and thus simulating data with an identical structure to that of the empirical analysis. After a decision is made in a period, consumers' utilities for products change as a function of the characteristics of past browsed products. More specifically, if product j was considered by the consumer in the last period,

then the utilities derived from products with similar characteristics to j are reduced in future periods. Thus, the product that a consumer browses or purchases in subsequent periods depends on the product they browsed or purchased in the previous period. Once these data are simulated, I estimate demand-prediction models identically to the empirical analysis. In this simulation, I also find that the JC model performs well in prediction relative to the characteristics model.

Related Literature

This paper contributes to the literature on learning demand-relevant latent characteristics, which are often computed through embedding algorithms (Rudolph et al. (2016), Liang et al. (2016), Barkan and Koenigstein (2016)). The effectiveness of these latent characteristics hinges on their capacity to accurately represent underlying consumer preferences. However, these characteristics may not adequately capture the impacts of items already in a consumer's basket, potentially failing to reflect true patterns of substitution and complementarity. This issue is particularly prevalent in online market settings where essential drivers of search like marketing influences are often unobserved. The Joint Consideration (JC) measure introduced in this study addresses this gap by providing a robust method for constructing latent characteristics that represent all consumers' substitution patterns effectively.

The JC measure could potentially be used as input to an embedding algorithm like t-distributed Stochastic Neighbour Embedding (t-SNE) (van der Maaten and Hinton (2008)), where distances in the embedding space might serve as latent characteristics in the vein of the aforementioned literature. However, due to the computational demands of processing tens of thousands of products, this approach is not feasible in my analysis.[3] Additionally, the computational complexity of such methods makes them impractical for applications requiring real-time predictions, such as in targeted advertising.[4] In an application with a smaller dataset, Magnolfi et al. (2022) use a measure similar to the JC measure as an input to the t-SNE method. They show that using these embeddings as latent characteristics results in a better fitting demand model than using only observed characteristics, when predicting aggregate shares in the ready-to-eat breakfast cereal market.

The revealed preference approach used to compute the JC measure for measuring substitution is closely related to the work by Armona et al. (2021), who also compute latent characteristics using search data by computing an embedding. Their embedding relies on a revealed preference approach; products that consumers search have higher utilities than products that were not searched resulting in a revealed preference inequality for each pair of searched and unsearched products. This approach could lead to dimensionality issues due in a setting with a many products, and in comparison the JC measure is very simple to compute.

[3] To help visualize the differences between using the JC and characteristics-space explanatory variables I use the t-SNE method (van der Maaten and Hinton (2008)) in the Methodology section. Due to its computational demands, I cannot apply the t-SNE method to the entire set of the observed products.

[4] For example, targeted product advertisements are decided by algorithms that take into account consumers' likelihood of purchases each time a consumer searches and must be completed in seconds.

Magnolfi et al. (2022) take a different empirical approach to compute embeddings by collecting survey data in which respondents reveal their beliefs about cereals they perceive as similar in the form of triplet comparisons, which they apply to a triplet embedding algorithm (van der Maaten and Weinberger (2012)). They use their computed embeddings as latent characteristics in a demand model, and show that using these latent characteristics outperforms using observed characteristics in predicting aggregate shares.

The use of repeated decisions of consumers in identifying substitution patterns can also be seen in the broader demand estimation literature. In the marketing literature, panel-data models of demand often include estimation of latent characteristics jointly with the parameters of the demand model (Elrod (1988), Elrod and Keane (1995), Keane (1997)). Identification of the latent characteristics, which in these models represent market structure, comes from consumers who switch from one product to another over time implying that they are substitutes. Since Berry et al. (1995), the characteristic-space approach has been more widely used in the empirical industrial organization literature, and relies on consumer preferences over product characteristics to determine substitution patterns. Within this framework, Berry et al. (2004) show how to use repeated purchase data to better identify substitution patterns. More recently, the marketing literature has devoted much effort in modeling consumer choices using search data, where a consumer's search process is explicitly modelled (De Los Santos et al. (2017), Ke and Villas-Boas (2019), Ursu et al. (2020), Moraga-González et al. (2023)).[5] In addition to the papers referenced above, Kim et al. (2011) develop a method to estimate latent characteristics using search data and in addition estimate preferences over these characteristics in a search model. As their results show, since most search models also follow the characteristic-space approach in modelling utility, the use of latent characteristics leads to improved model fit. My paper combines these ideas by estimating an easy-to-compute measure using search data that represent consumers' underlying substitution patterns and are explicitly included as explanatory variables (latent characteristics) in a predictive demand model.

A few papers have used measures similar to the JC measure in other contexts. Kumar et al. (2020) apply a heuristic similar to the JC measure in order to study product bundling. Using a clickstream dataset in which they observe users' shopping baskets they identify complements and substitutes using products purchased together and products considered but not purchased together respectively. Note that the latter idea is used in the definition of the JC measure. They use this method of categorizing complements and substitutes to augment an embedding algorithm and study how to optimally bundle products using the embedding representations of each product. In my work, the JC measure captures substitutability in precisely this mechanism. In comparison to their work, I show that in a predictive model of purchase the JC measure effectively captures substitutability between products. In addition, I use the measure as an input to a clustering algorithm rather than an embedding algorithm, which is more feasible in my empirical setting with almost 60,000 products. Atalay et al. (2023) use the likelihood of a household ever purchasing both of a pair of products in the Nielsen market basket panel as an input to an agglomerative clustering method.

[5] This is a rapidly expanding literature. The interested reader should see Honka et al. (2019) for a survey of the search and consideration set literature and Ursu et al. (2023) for a recent survey of the sequential search literature.

They then treat each computed cluster as a nest within a nested-logit demand model. To the best of my knowledge, my paper is the first to use this kind of repeated co-search measure for the purpose of a predictive demand model that takes past purchase behavior into account.

In addition to Atalay et al. (2023), other models in the emerging flexible demand estimation literature employ search data to better inform substitution patterns. Donnelly et al. (2024) and Amano et al. (2022) estimate a two-step consideration-then-choice demand model. The former paper estimates latent factors of utility that drive substitution while the latter constrains the correlations in a component of utility to rationalize product co-search patterns. Dotson et al. (2018) and Dotson et al. (2024) instead estimate a multinomial probit choice model and allow the utility covariance structure to depend on the similarity between products.

In order to evaluate the efficacy of the JC measure, I use it for explanatory variables in a demand prediction model and observe the resulting demand prediction performance. A number of papers have investigated models that fit and perform well in predicting demand. Bajari et al. (2015) apply several methods to estimate demand for salty snacks from panel data for a grocery store chain. They compare the fit of these methods as measured by RMSE and find that machine learning methods such as Random Forest show substantially superior predictive accuracy compared to traditional statistical methods such as linear/logistic regression. For brevity, I use just two methods for each choice of explanatory variables; logistic regression and a very closely related algorithm to the Breiman (2001) Random Forest algorithm - LightGBM developed by Ke et al. (2017). LightGBM and Random Forest are both aggregated tree-based methods but differ in the way trees are constructed. LightGBM uses a more computationally feasible method in constructing trees, which works well given that I use a very large dataset. I describe the details of this method in the Methodology section. I show that the effective performance from using the JC measure holds for both estimation methods. I also evaluate the models using two common measures of model fit in the machine learning literature; the Area Under the Curve (AUC) of each model's Receiver-Operating-Characteristic (ROC) plots and an aggregated Cross-Entropy Loss[6]. Both measures of fit suggest the same ranking of models as the misclassification rate.

There is a vast literature in predicting consumer choices for eCommerce. As clickstream data has become more accessible to researchers, a number of recent papers have developed predictive demand models that can use consumers' past search choices to predict purchases. Recently in the marketing literature, such models have been developed by improving upon existing machine learning methods (Gabel and Timoshenko (2022), Jacobs et al. (2016)). These recent models allow for consumer preferences over goods in different categories by computing latent characteristics, often using embeddings, to represent all products in one characteristic-space. The latent characteristics estimated using the JC measure in this paper accommodates for products in different categories to affect purchase probabilities in this way. Earlier models often included additional parameters to capture cross-category relationships (Manchanda et al. (1999), Russell and Petersen (2000)).

[6]Both measures of fit are explained in the Methodology section.

Ruiz et al. (2020) develop a model to predict the next item a consumer will choose for their shopping basket given the set of items already chosen. The effect of products already in a basket on future choices is modelled through a term capturing the effect of characteristics of products already in the basket. In contrast, the model of Shiller (2020) measures the impact of past behavior on the probability of purchase using product fixed effects.

While not directly focusing on price discrimination, this paper highlights the potential of using clickstream data to develop predictive demand models, which is vital for strategies like personalized pricing. Dubé and Misra (2023) develop a model of personalized pricing by estimating a demand equation using machine learning methods to deal with dimensionality issues. Shiller (2020), Smith et al. (2023), Waldfogel (2015), and Zhang et al. (2014) estimate individual level demand as a function of consumers' observables and past purchase behavior, and find large improvements in predicted revenues particularly when taking past purchase behavior into account. By demonstrating the capability to extract demand-relevant latent characteristics from clickstream data alone, this paper provides valuable insights for online retailers looking to optimize demand prediction without extensive product characteristic data.

Data

I use a substantial clickstream dataset detailing consumer product browsing behaviors on a UK eCommerce store during October and November 2019, available via the "Kaggle.com" data repository (Kechinov (2019)). The specific identity of the eCommerce platform is undisclosed; however, a histogram of product categories, shown in Figure 6 in the Appendix, provides insight into the platform's product offerings. The dataset encompasses detailed records of user interactions with product pages, classifying each interaction as one of four event types: purchase, view, add-to-cart, remove-from-cart. Additionally, product characteristics data such as price, main-category and sub-category labels, and brand are included. The latter three are used as discrete characteristics, and price is used as a continuous characteristic.

Table 1: Variables in Dataset

Variable	Description
event_type	One of: view, cart, remove from cart, purchase
event_time	Time of event
category_code	Name of main and sub category
brand	Brand name
price	Price of product
user_id	Permanent user id

The data are then transformed such that information from past events is assigned to each current event. Specifically, I assign to each event the information of the past l events from that user. Consider the following example observations in Table 2.

Table 2: Example of Raw Data

event_type	category_code	brand	price	event_time
View	computers.desktop	P	900	2019-10-01 00:00:05
Purchase	computers.notebook	L	700	2019-11-01 00:00:01

Here the example user first viewed a desktop from brand "P" and then one day later purchased a notebook from brand "L". Suppose that $l = 1$ so that each observation must be assigned one previous event. The first observation is dropped since it is the first event in the data for this user. The second event will be assigned information from the first event. The resulting observation in the transformed dataset is given in Table 3.

Table 3: Example of Raw Data - Transformed

Purchase	Price	Main_category_same	Sub_category_same	Brand_same	Event_type	Price_diff
1	700	1	0	0	View	200

The outcome variable, "Purchase", is an indicator that "event_type" takes the value "purchase". Thus, the goal of all models is to classify purchase of the product corresponding to this outcome variable, hereafter referred to as the outcome product. The information from the most recent event is summarized in the latter six variables. The transformation assigns values indicating whether the discrete product characteristics (main-category, sub-category, and brand) match those of the outcome product, and the difference in price with the outcome product thereby measuring distance in observed characteristics-space. Dummy variables for event type of the past product will be interacted with each of these distance variables. This set of six variables is repeated for each of the previous l events assigned to the outcome event.

Choosing l involves a trade-off between the inclusion of sufficient past events to enhance model accuracy against the risk of reducing the sample size. As l increases there are more previous events' information to use to predict purchases and generally this should lead to an improvement in model fit and prediction performance. However, this results in fewer observations for which we observe at-least l previous events for a user, making the models less accessible in predicting purchases for users newer to the platform. Choosing l while considering this trade-off is thus subjective. I opt for $l = 10$ as a conservative balance to maintain robustness across a wide range of user behaviors while performing sensitivity analysis for l values between 5 and 13. Each increase in l corresponds to a decrease of roughly $200,000$ observations, or around 2% of all observations.

A natural challenge with clicksteram data for predictive modeling is that some observations capture users at a late stage in the buying process. Typically, a user's search history will show a view followed by an add-to-cart event shortly before a purchase. Directly using this sequential information as explanatory variables can increase the model's prediction accuracy; however, it does not translate effectively into real-world applications. For instance, if this model were used in real-time to trigger product advertisements,

these would likely reach users who have already decided to buy the product and are merely completing the necessary purchasing steps. This scenario would render the advertisements redundant, potentially wasting marketing resources. To circumvent this limitation, I ensure that the first event in the sequence of l events used to construct the explanatory variables is associated with a different product than the outcome event.

Methodology

Predictive-demand models play a crucial role in marketing by leveraging historical consumer behavior to forecast future purchases. For instance, marketers often aim to predict which product j a consumer i is likely to purchase, based on a set of products J they have previously searched for the purpose of targeted advertising or product recommendations. In this paper, all models are designed with the goal of using the historical search sets J to predict the likelihood that consumer i will purchase a currently browsed product j. As mentioned previously, search sequences in the dataset conclude with the purchase of one product, suggesting that the products within each sequence are likely substitutes. Therefore, the explanatory variables are constructed to quantify the substitution patterns of each previously searched product j in J on the outcome product j. Typically, this involves using distance measures in characteristic-space, where the probability of purchasing the outcome product j is modeled based on the similarity of characteristics between each j and j. This kind of model serves as the benchmark characteristics model in this paper.

However, this paper explores an innovative approach by developing an alternative substitution measure that captures consumers' revealed substitution patterns derived from their product search patterns. The construction of this novel Joint Consideration (JC) measure, alongside traditional characteristics-space measures of substitution, and their application in predictive models are detailed in the following sections. This approach not only leads to high predictive accuracy but also provides deeper insights into how substitution patterns can be learned from consumer search patterns.

Joint Consideration

The primary contribution of this paper is the development of the "Joint Consideration" (JC) measure, a novel metric designed to quantify the substitution relationships between products based on aggregated consumer search patterns. This measure leverages the entire dataset to crowdsource a broad understanding of consumers' perception of substitute goods. The rationale is that if products j and j' are frequently in the same search sets among all users in the data, then many users perceive them as closely related products. So when predicting whether a consumer will purchase j, having observed the user browse j' on the platform already should be informative of the choice the user will make when considering j.

A search set for each consumer is defined as the collection of products viewed before a purchase decision is made. The JC measure is calculated by first identifying all observed search sets, which include any products associated with "view" events within a user's search sequence. The dataset's structure ensures that the

purchased product is also tagged as viewed, thereby including it in the search set. Events subsequent to a purchase are treated as part of a new search sequence for that user.

The JC measure for a pair of products j and j' is given by the following:

$$JC(j, j') = \frac{\#\text{ search sets with } j, j'}{\#\text{ search sets with } j \text{ or } j'} \tag{1}$$

Once computed, this measure is added as an explanatory variable for each pair of current and past event. Returning to the example of data in Table 3, if it is calculated that the JC measure between the desktop and notebook is 0.5, then the transformed data become as in Table 4 below.

Table 4: Example Transformed Data

Purchase	Price	Main_category_same	Sub_category_same	Brand_same	JC	Event_type	Price
1	700	1	0	0	0.5	View	900

There are 57,048 products in the data and so if too few search sets are observed, the matrix of JC measures will be sparse. Thus, this measure is constructed using the set of observed search sets from roughly half of the data; the first month of data from October. More precisely, only search sequences that end in October are used. The models which use the JC measures are then restricted to be estimated on only data from November. This also ensures that data from the latter month are not used to construct the measure which in turn is used to predict purchases from the same month, which cannot occur in practice. The characteristics model which does not use the measure is allowed to be estimated on the data from both months. This allows a fair comparison in the sense that the models which do not use the JC measure do not need to set aside a portion of the data to construct an explanatory variable.

Table 5: Search Sequences Summary Statistics

Statistic	Mean	Std	25p	50p	75p	90p	Min	Max
Number of Events	7.67	12.62	3.00	4.00	8.00	16.00	1.00	933.00
Number of Products	3.39	5.37	1.00	2.00	3.00	7.00	1.00	301.00
Duration (Hours)	18.32	37.33	0.02	0.10	16.52	73.81	0.00*	168.00

* 139,741 sequences only consist of one search or one purchase and have a duration of 0.

Table 5 provides summary statistics on the 1,211,593 observed search sequences from both months of data. Search sequences are typically short; the 75th percentile of the number of events in a sequence is 13, and is 9 for the number of products in a sequence. Most search sequences end in a short time period, with the 50th percentile for search duration at 2.44 hours.

Table 6: JC Measure Summary Statistics

Statistic	Mean	Std	25%	50%	75%	90%	99%	Min	Max
Unconditional JC	0.007	0.021	0.001	0.002	0.006	0.015	0.071	0.000	1.000
Within Search Set	0.010	0.023	0.001	0.002	0.007	0.027	0.115	0.000	1.000
Within Search Set (Weighted)	0.023	0.041	0.002	0.009	0.029	0.063	0.160	0.000	1.000

Table 6 provides summary statistics for the JC measure. The first row gives summary statistics for the JC measure for all possible pairs of products. The second row instead restricts the pairs to those that appear at-least once in a search set. Finally, the third row weights the statistics by the frequency with which each pair of products appears in a search set. The JC measure for products within a search set typically are higher than for any two products. Thus, after observing a user's browsing history, it is sensible that a product with a high JC measure is more likely to be purchased than those with a lower JC measure.

Visualization of Substitution Patterns

To visualize the differences between substitution measures constructed from either observed characteristics or the Joint Consideration (JC) measure, I apply the t-distributed Stochastic Neighbour Embedding (t-SNE) algorithm (van der Maaten and Hinton (2008)). This algorithm takes a distance measure as an input, and computes a low-dimensional vector representation of each product such that the distances of products in the low-dimensional space reflect the inputted distance measure, thereby reducing the dimensionality of data while maintaining the relative distances between points.[7] When plotted, the embeddings often reveal clusters in high-dimensional data, making it ideal for visualizing complex relationships like product substitution (van der Maaten and Hinton (2008)).

Due to its computational intensity, with time and memory complexity scaling as $O(n^2)$, I constrain this analysis to a randomly selected subset of 10,000 products. To further aid in visual interpretation and highlight the product substitution structure in the data, I apply the Partitioning-around-Medoids (PAM) clustering algorithm to group close products. The number of clusters, set to 6 and 8 for each of the two substitution measures based on silhouette analysis, helps identify groups that are close in characteristic-space or have high JC measures.

The observed characteristics are a mix of qualitative and quantitative variables. Therefore, I use Gower's measure presented in Gower (1971) to find the distance between two products as it can accommodate both qualitative and quantitative data as characteristics. If each product has p characteristics indexed by a, the distance between products $x_j = (x_{j,1}, ..., x_{j,p})$ and $x_k = (x_{k,1}, ..., x_{k,p})$ is given by the following:

[7] For complete details of the method, the interested reader should see van der Maaten and Hinton (2008).

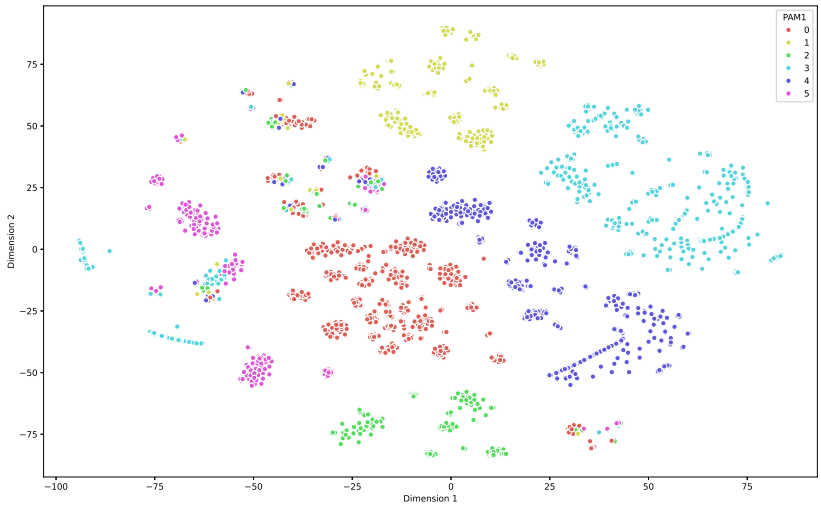

Figure 1: Observed characteristics

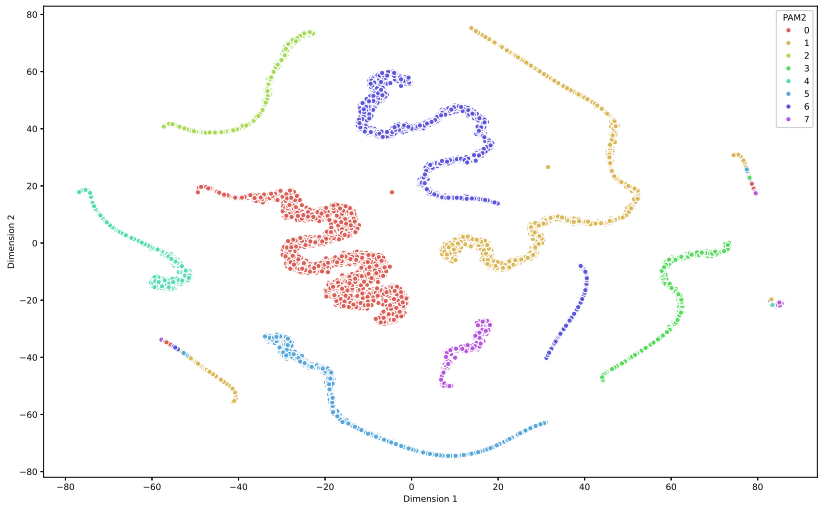

Figure 2: JC Measure

$$D_{\text{Observed}}(j,k) = \frac{1}{p}\sum_{i=1}^{p} d_a(x_j, x_k), \tag{2}$$

$$d_a(x_j, x_k) = 1(x_{j,a} \neq x_{k,a}) \text{ when } a \text{ is qualitative}, \tag{3}$$

$$d_a(x_j, x_k) = \frac{|x_{j,a} - x_{k,a}|}{R_a} \text{ when } a \text{ is quantitative}, \tag{4}$$

where R_a is the range of characteristic a in the data. The distance matrix when using the JC measure is given by

$$D_{\text{JC}} = 1 - JC \tag{5}$$

Visual inspection of the positions of points representing each product in Figures 1 and 2 illustrates distinct differences between the two distance measures used. Figure (1) uses D_{Observed} to plot embeddings and color-codes products according to their respective clusters. In contrast, Figure (2) uses D_{JC} for its embeddings and clusters. As discussed previously, each of the two distance measures aim to identify substitute products as those with a small distance, which in these Figures will be represented by points close in the two-dimensional Euclidean space. Thus, any differences in identifying substitutes for each distance measure will be revealed by visual differences in each two-dimensional embedding.

Two key differences between using D_{Observed} and D_{JC} are apparent. First, D_{JC} results in clearly defined clusters where products within a cluster are tightly grouped, and there is a significant separation between different clusters. Accordingly, the products that should be within a cluster are much more apparent when using the JC measure. Such clarity in clustering aligns with expectations in a multi-category market where consumers perceive certain groups of products—whether from different categories or within a single category like footwear (encompassing varieties such as running shoes and boots)—as substitutes. On the other hand, D_{Observed}, as depicted in Figure (1), produces a more scattered plot, suggesting that traditional characteristics may not capture subsstitution patterns as effectively as the JC measure.

Comparison of Models

The effectiveness of the JC measure is assessed by comparing prediction performance for the characteristics model that uses observed characteristics to the JC model that does not. Specifically the following sets of explanatory variables, all interacted with event type dummy variables, are used in the different models for each of the l pairs of j and j':

Table 7: Sets of Explanatory Variables

Model Label	Explanatory Variables
Characteristics	Identical Category/Brand Indicators, Difference in Prices
JC	JC measure
Characteristics plus JC	Characteristics Variables, JC measure

The 'Characteristics' model uses distance in characteristic-space as a measure of substitution, while the 'JC' model focuses solely on the substitution relationships inferred from search sequences. The combined 'Characteristics plus JC' model integrates both approaches, since both measures are likely informative of substitution patterns.

Table 8: Conditional Means of Explanatory Variables

	Outcome of Previous Event	
Variable	Purchase	Nonpurchase
Main Category	0.604	0.681
Sub Category	0.535	0.622
Brand	0.765	0.470
JC	0.358	0.573

The proportion of purchases is 0.065 and the mean price is £374.82.

Table 8 presents mean values for the explanatory variables, conditioned on the outcome of the previous event. For example, if the most recent event is a purchase, the outcome product shares the main category with the most recent event's product in 60.4% of observations, compared to 68.1% following non-purchase events. Thus, as consumers are browsing products, they substitute away from products in the same main category more aggressively after a purchase. The same pattern is true for the subcategory of the product and the JC measure. However, the brand of the product exhibits the opposite trend, suggestive of a persistent brand value effect. This suggests a pattern in search behavior: once a satisfactory product within a category is purchased, search often shifts to a different product type. Equivalently, consumers tend to browse products close in characteristic-space before making a purchase decision. In summary, the most recent events tend to correspond to products that share characteristics with that of the outcome variable, and changes in search behavior depend on whether the consumer has decided to make a purchase. This supports the use of substitution measures as explanatory variables in each model.

Comparing these search patterns to the summary statistics in Table 6, we see that products in consecutive events have JC measures well above the 99th percentile of its distribution. This further suggests that, as expected, users browse close substitutes in their search sequences, and so the JC measure effectively captures this pattern.

Estimation

Each method estimates the probability of the outcome product's binary purchase decision being equal to 1 given the last l searches, as a function of the explanatory variables given formally by the following:

$$P(\text{Product } j \text{ purchased}|J) = f(x_j, p_j, I(x_{j'} = x_j)E_{j'}, (p_{j'} - p_j)E_{j'}, JC_{j'}E_{j'}) \tag{6}$$

The function $f()$ is linear in each of its inputs, with the explanatory variables varying in each model as given in Table 1. The specific estimating equation for logistic regression is given in equation (7). Apart from price and the JC measure (and their interactions with event type dummy variables), the explanatory variables are all binary variables. Both logistic regression and LightGBM estimation methods handle these

well with the use of regularization, which I discuss next.

Logistic Regression

Since the objective is prediction, or more precisely classification, for both types of models, we must also deal with the issue of overfitting. Logistic regression models in particular are prone to overfitting when the number of parameters is high (Hastie et al. (2001)). I employ L2 regularization, also known as ridge regularization, to deal with this issue. L2 regularization applies a penalty term given by the size of the estimated parameters which has the effect of shrinking the parameters closer to zero, thus reducing the number of explanatory variables that have a large impact on the prediction. Thus, it becomes less likely that irrelevant explanatory variables can be used to greatly influence predictions and thus overfit the model. The logistic regression equation is given by the following:

$$\log \frac{p}{1-p} = x_j\beta + p_j\alpha + \sum_{j'}[I(x_{j'} = x_j)E_{j'}\beta' + p_{j'}E_{j'}\alpha' + JC_{j'}E_{j'}\gamma'] \tag{7}$$

Let θ be the vector of parameters $(\beta, \alpha, \beta', \alpha', \gamma')$. Then the L2 regularization technique chooses parameters by solving the optimization problem given by the following:

$$\hat{\theta} = \arg\min_{\theta} \quad -log(L) + \lambda \sum_i |\theta_i|^2, \tag{8}$$

where λ is a tuning parameter that controls the importance of the L2 penalty term. A higher value of λ corresponds to a more aggressive penalty. I choose the value of λ that minimizes the misclassification rate in the validation sample by grid search over the values $\{0.1, 0.2, ..., 1\}$ independently for each logistic regression model.

Gradient Boosted Random Forests

LightGBM, developed by Ke et al. (2017), is a very similar estimation procedure to the more popular random forest model of Breiman (2001). The LightGBM model differs in the estimation of a classification tree f_b in step 2 (b) below and the use of gradient boosting. In the Breiman (2001) random forest model, estimation of classification trees is based on the CART algorithm. In the LightGBM model, estimation is based on a "leaf-first" method. The leaf-first method is much cheaper computationally, and generally leads to better results. Each tree is also estimated through a gradient boosting procedure, which the literature has shown to work better than the simpler Random Forest model (Hastie et al. (2001)). The interested reader should see Breiman (2001) and Ke et al. (2017) for details on these tree estimation methods.

I briefly describe the LightGBM algorithm below.

1. Sample, with replacement, B training samples (y_b, X_b).

2. For each b in B

   (a) Randomly sample from the set of predictors X_b.

   (b) Estimate a classification tree f_b on sample (y_b, X_b).

3. The final prediction is given by $\hat{f} = 1(\frac{1}{B}\sum_{b=1}^{B} f_b(y_b, X_b) \geq 0.5)$, or in other words majority vote.

In addition to the evidence in the literature (Bajari et al. (2015)) that random forest models work well in predicting demand, they are typically robust to over-fitting issues (Hastie et al. (2001)). Since the set of predictors is randomly sampled in each tree, and the predictions of the trees are aggregated, there is a low chance that any tree which overfits the data to a set of irrelevant variables significantly influences the aggregated prediction. The same applies to the LightGBM model as well.

The LightGBM model has a very large number of tuning parameters. I use the "LightGBM Regressor" from Python's "LightGBM" package. Following the recommendations of Ke et al. (2017), I set the majority of tuning parameters to the default values, but increase the number of trees estimated, B, to 500 and decrease the learning rate to 0.01. Both these changes have the effect of increasing the effectiveness of the predictions. For a full discussion of all tuning parameters, see Ke et al. (2017).

For all methods, I randomly sample 50% of observations as the training sample, 25% of the observations as the validation sample, and the remaining 25% of observations are used as the test sample. The validation sample is used to assess the sensitivity of the results to the choice of the number of clusters k and to choose λ for the logistic regression model.

Results

Both the logistic regression and LightGBM models give predicted probabilities of purchase.[8] In order to classify the outcome as a purchase/non-purchase, a probability threshold must be chosen so that a predicted probability above the threshold is classified as a purchase and vice versa. These classifications can be assessed by calculating the corresponding False Negative Rate (FNR) and False Positive Rate (FPR). The FNR and FPR, or Type I and Type II error rates, are the proportion of purchases misclassified as non-purchases and the proportion of non-purchases misclassified as purchases, respectively.

To assess prediction performance, I use a number of common measures of model fit from the prediction literature. First, each model is assessed using the receiver-operating-characteristic (ROC) plot, which assesses the predicted classes of the model. After calculating predicted classes for many choices of thresholds between 0 and 1, an ROC plot plots the combinations of the False Positive Rate and the True Positive Rate (1-False Negative Rate) for each set of predictions. A practitioner would have to choose one probability threshold in order to obtain a prediction that can be used for decision making, but the true cost of each type of false

[8] LightGBM can be used for binary classification directly as well. For the sake of comparison with the logistic regression model, I use the LightGBM regressor instead as this allows for the calculation of ROC curves and Cross-Entropy Loss.

prediction must be considered and depends on the application. As this is unknown to the researcher, the ROC plot can be used to remain agnostic about the choice of threshold for classification and assess model performance. A curve at the 45-degree line implies that the model holds no predictive value. A curve above this line implies a better fitting model. Then the area under the curve (AUC) is a measure of fit; an AUC of 0.5 implies no predictive value and a higher AUC implies a better fitting model. The ROC curves for each model and their corresponding AUC are presented in Figures 3 and 4.

Table 9 also reports the results for the AUC of each model, in addition to another common measure of model fit named the Cross-Entropy Loss. The Cross-Entropy Loss of an individual observation i, CEL_i, is equivalent to the log-likelihood equation and is given by (9). This measure indicates how well the predicted probabilities approximate the underlying binary purchase decisions.

$$CEL_i = y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i) \tag{9}$$

Columns 2 and 3 of Table 9 report the AUC for each model over the test sample and Columns 4 and 5 report the mean Cross-Entropy Loss across all observations in the test sample.

Table 9: AUC and Cross-Entropy Loss

	AUC		Cross-Entropy Loss	
Model	Logistic Regression	LightGBM	Logistic Regression	LightGBM
Characteristics	0.8928	0.9053	0.1847	0.1895
JC	0.8837	0.8966	0.1885	0.1848
Characteristics plus JC	0.9047	0.9125	0.1799	0.1825

The main result is given by comparing the JC model to the Characteristics model. The JC model performs comparably to the Characteristics model, and using the JC measure actually leads to an improvement in fit when used in conjunction with the characteristics-based explanatory variables. Complementing the results from Shiller (2020) who uses a fixed-effects approach[9] to determine the impact of past browsing behavior on purchase probability, I find a larger AUC for the Characteristics model (0.9053 vs. 0.7100), suggesting that explicitly measuring the relation using substitution measures between j and each j' is important for prediction when taking into account past behavior.

I view the fact that the JC model performs comparably to the Characteristics model as evidence of its effectiveness. Recall the pattern of the summary statistics in Table 8 that products in the most recent events have relatively high JC measures with the outcome product. In combination with these results, they suggest that in predicting the purchase of product j, observing the user having browsed products which are close substitutes as per the JC measure is informative of consumers' purchase decisions. Thus, the JC measure alone effectively captures the degree of substitution between products, and as well as demand-

[9]Their model predicts Netflix subscription status using variables indicating whether the consumer ever visited and made a purchase at each other website observed in their data. See Shiller (2020) for details.

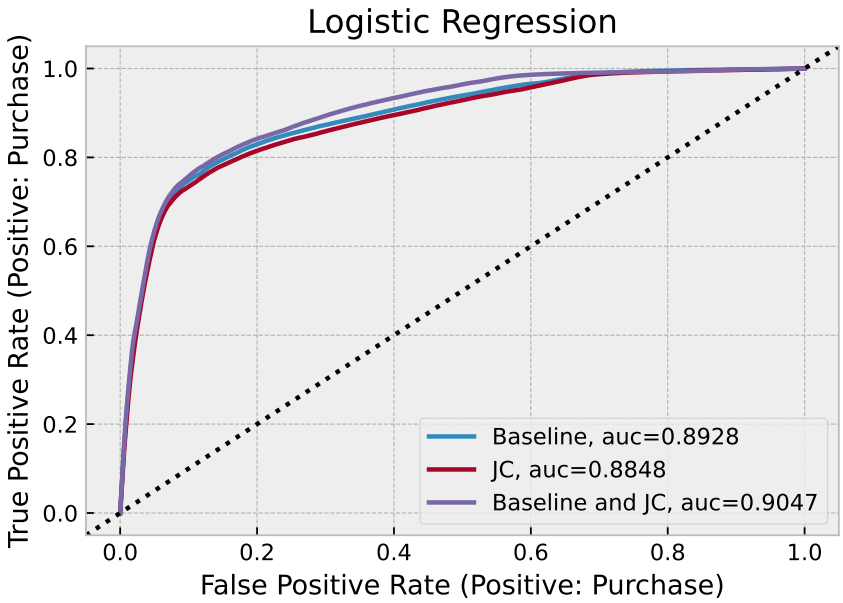

Figure 3: ROC Curves for Logistic Regression Models

relevant characteristics. Since the JC model performs comparably, if characteristics data were not available to construct explanatory variables, then these results suggest that inferring substitution patterns using the JC measure is effective, and can be used to build an effective predictive model.

To gain an intuitive understanding of the accuracy of predictions, I present the misclassification rates for one choice of threshold. As is common in the machine learning literature, I choose the probability threshold that maximizes the Youden (1950) J-statistic:

$$J = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}} + \frac{\text{true negatives}}{\text{true negatives} + \text{false positives}} - 1 \tag{10}$$

For this set of classifications, I present the overall misclassification rate as well the FNR and FPR. First, the Characteristics model performs well in predicting purchases, achieving an overall misclassification rate of 13.59% and 12.79% for each estimation method. The JC model also performs well, and in fact gives a small improvement relative to the Characteristics model for logistic regression.

Table 10: Purchase Misclassification Rates (%)

	Logistic Regression			LightGBM		
Model	FNR	FPR	Mis	FNR	FPR	Mis
Characteristics	22.34	12.78	13.59	20.70	12.05	12.79
JC	25.18	11.35	12.53	20.79	12.45	13.16
Characteristics plus JC	20.78	13.29	13.92	20.42	11.40	12.17

The threshold probabilities for the logistic regression and LightGBM models are 8.16% and 9.45% respectively.

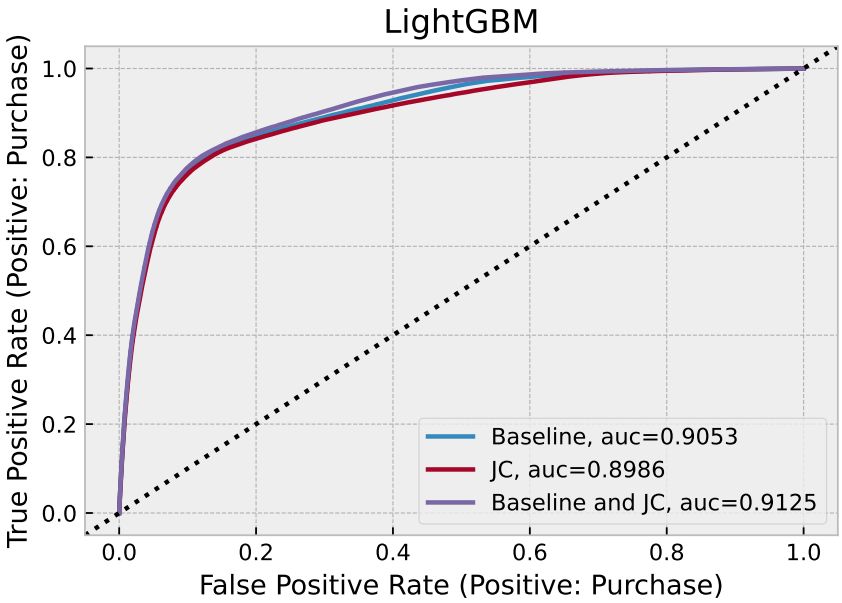

Figure 4: ROC Curves for LightGBM Models

In this empirical example, I constructed the JC measure using one month of data. There are 57,048 products in the sample, and thus the total number of JC measures to compute is $57,048^2 \approx 3.25$ billion. In the search sets ending in the first month of the sample, we observe roughly 50 million pairs of products across all search sets. In principle the JC measure's improvement of prediction performance should scale with more past search data available, which is likely the case in industry applications. To demonstrate its scalability, I recompute the JC measure using only the search sets ending within the first two weeks of the sample period. When computed using this sample, we observe roughly 3 million pairs of products in these search sets. Accordingly, the results in Table (11) show that there is a reduction in prediction performance, suggesting that the JC measure scales well when more data are available. For an online retailer with much more than two months of historical search data, the JC measure will likely show improved performance since more of the possible pairs of products in the JC matrix would be observed in search sets.

Table 11: AUC and Cross-Entropy Loss

	AUC		Cross-Entropy Loss	
Model	Logistic Regression	LightGBM	Logistic Regression	LightGBM
JC	0.8837	0.8966	0.1895	0.1854
Characteristics plus JC	0.9036	0.9117	0.1804	0.1840

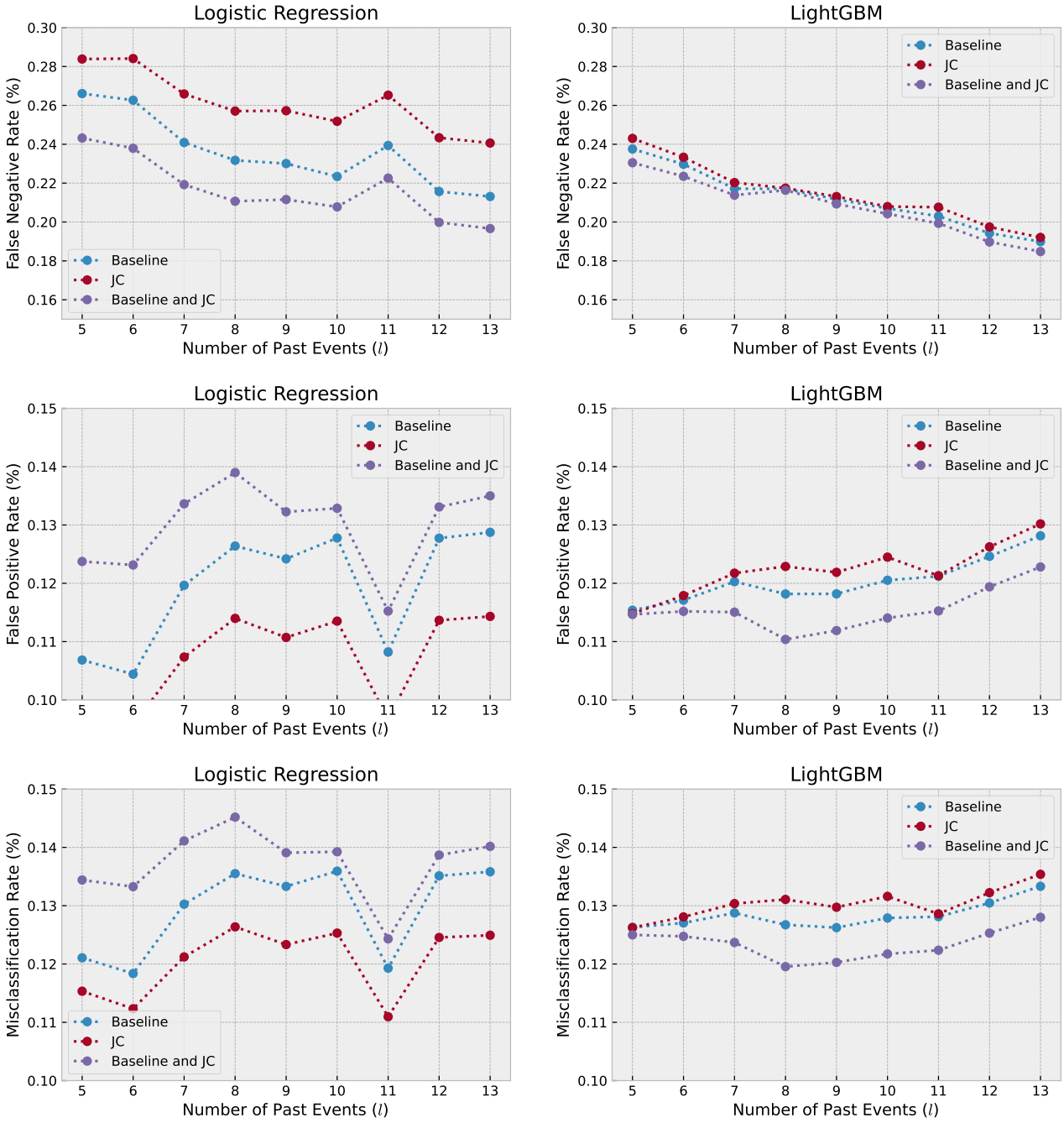

Figure 5: Results by choice of $l = 5 - 13$

Sensitivity Analysis on Choice of l

The main results are for a choice of $l = 10$. Figure 5 presents the misclassification results for all models when l is varied from 5 to 13. The results are presented for the FNR, FPR, and misclassification rate. There are two key findings in these results. First, the performance of all models generally see an improvement in performance as l increases. Second, the relative performance of the models do not change drastically as l varies. The JC model continues to perform comparably to the Characteristics model as l increases. False positive rates seem to worsen as l increases, nevertheless the rankings of the model are fairly consistent. Thus, the results seen in the empirical analysis are robust to changes in l. Each increase in l is associated with a loss of roughly 200,000 observations. Thus, I make the subjective decision to set l to 10. Further increases in l will result in fewer observations,[10] and as Figure 5 suggests, there will not be a significant increase in performance.

Monte Carlo Simulation

As a second check on the effectiveness of the JC measure in the predictive models, I conduct a similar analysis as before using simulated data. In the simulations users purchase an available product or the outside good each period. In order to simulate search, each individual's preferences for products in the future changes as a function of the characteristics of past searched products. The researcher observes each individual's decision on which product is purchased today and also view four historical purchase decisions. Precisely how preferences change over time is calibrated from the data in the main analysis, which I discuss below.

In each replication, I generate 10,000 observations of individuals' purchase decisions in five periods, denoted j_t. There are 64 products in the product space, each assigned one of four main-categories (mc), one of four sub-categories (sc), one of four brands (b), and one of 16 unobservable characteristics (u). Consumer i's utility for product j in period t is given by

$$U_{it}(j) = \alpha_{it}^{mc} + \beta_{it}^{sc} + \gamma_{it}^{b} - p^{j} + u_{it}^{mc,b}.$$

Here α_{it}^{mc} is the utility to consumer i in period t of purchasing a product within main-category mc. Similarly, β_{it}^{sc} is the utility of sub-category sc, and γ_{it}^{b} the utility of brand b. The utility from the unobservable characteristic $u_{it}^{mc,b}$ depends on the main-category and brand. Thus, it can be thought of as the additional utility of the product being produced by brand b specifically in the main-category mc. This is important for the change in preferences between periods discussed below. Consumers receive a deterministic income y_i each period, and can purchase a product in each period if they have accumulated enough income. The consumer can always purchase the outside option without paying a price, which will be interpreted as the

[10] A model that can be estimated on fewer observations is less applicable in practice. For marketing applications for example, fewer consumers can be targeted with advertisements when the model requires omission of observations for consumers with short search histories.

Table 12: Conditional Means

	Simulation		Main Analysis	
Variable	Purchase	Nonpurchase	Purchase	Nonpurchase
Main Category	0.375	0.461	0.604	0.681
Sub Category	0.292	0.386	0.535	0.622
Brand	0.472	0.373	0.765	0.470
Joint Consideration	0.042	0.085	0.358	0.573

researcher observing that the consumer browsed the highest utility product in the product space, but did not purchase it. The consumer's problem is to choose products j_1, j_2, j_3, j_4, j_5 to purchase (including the outside option) in each period to solve the following:

$$\max_{j_1,j_2,j_3,j_4,j_5} \sum_{j_t} U(j_t) \text{ subject to } p_{j_t} \leq \sum_t y_i - p_{j_{t-1}} \text{ for all } t. \tag{11}$$

In order for the individual's search and/or purchase choices to change between periods, there must be a change in the utilities from the most desirable - largest utility - product in the space. Specifically, if product j with characteristics (mc, sc, b) is purchased in period t, then in period $t+1$ one or more of $\alpha_{it}^{mc}, \beta_{it}^{sc}, \gamma_{it}^{b}$ must change. The percentage change in each is calibrated such that the DGP results in similar summary statistics as compared to the main analysis in Table 8. After calibration, I set that the change in each is $-15\%, -15\%, 0\%$ respectively. That is, the utility from categories (mc,sc) are reduced by 15%, while the utility from the brand does not change. The DGP summary statistics are given in Table 12. Intuitively, users substitute away from goods in the same categories as they have bought, but stick within a brand. The utility from the unobservable characteristic does not change either, which can be thought of as the complementarity of all goods produced by brand b in each main-category. The pattern in the DGP summary statistics is largely consistent with that of the main analysis.

This simulation also allows us to verify that the JC measure is capturing substitutability between goods that comes from information unobservable to the researcher. The characteristics in the utility DGP are all observed apart from the u_{it} term. Thus, the JC model can only perform well relative to the Characteristics model if it effectively captures the degree of substitution between products with similar values for this unobservable characteristic. This would also imply that the Characteristics and JC model should lead to an improvement relative to the Characteristics model.

Finally, the simulation allows for the calculation of confidence intervals. The results in Table 13 present the mean fit metrics, as well as confidence intervals using the 2.5th and 97.5th percentile of each fit metric across all 10,000 replications.

In this simulation I assess the performance of the logistic regression model and the LightGBM model as in the previous section. Here the models predict purchase probabilities for product j_5 given j_1, j_2, j_3, j_4. The results are summarized in Table 13.

Table 13: Misclassification Rates (%) and Area Under the Curve

	Logistic Regression			LightGBM		
Model	FNR	FPR	AUC	FNR	FPR	AUC
Characteristics	2.59	22.54	93.34	1.96	18.74	95.30
	[2.58,2.60]	[22.47,22.65]	[93.31,93.34]	[1.90,2.96]	[17.70,18.81]	[95.28,95.32]
JC	5.48	19.23	92.26	1.75	20.69	93.70
	[5.45,5.61]	[19.20,19.30]	[92.23,92.28]	[1.67,1.79]	[20.67,20.82]	[93.58,93.73]
Characteristics and JC	4.51	20.46	93.50	2.57	18.00	95.37
	[4.49,4.86]	[20.12,20.50]	[93.49,93.52]	[2.35,2.67]	[17.76,18.14]	[95.36,95.43]

α, β, γ are drawn from $U(0,1)$, prices are drawn from $Beta(2,5)$, income is drawn from $Beta(1,50)$.

The number of replications is 10,000. The simulation results exhibit the same patterns as in the main analysis. The JC model performs comparably to the Characteristics model across all fit metrics, and the Characteristics and JC model leads to an improvement in performance. The confidence intervals are quite narrow, suggesting that these differences in performance do not arise due to noise. Thus, these results support the findings of the empirical analysis.

Conclusion and Managerial Implications

In this paper I present a method to extract substitution patterns of consumers using product search data. I construct a measure, called JC, of substitution between products by aggregating consumers' search sets, effectively crowdsourcing information on substitution patterns. To demonstrate its utility, I use the measure as an explanatory variable in a predictive demand model in which users' product search choices affect future purchase decisions. I show that by effectively capturing the degree of substitution between two products relative to characteristic-space measures, the JC measure alone can be used to effectively predict purchases even when product characteristics are not available. I also find that prediction performance is highest when both measures are used together.

To support these findings, I carry out a Monte Carlo simulation procedure in which users sequentially browse through products. Each product browsed is compared to the outside good and a purchase decision is made. Then, applying the same methods as in the empirical analysis, I show that the results in the simulation are consistent with those of the empirical analysis.

Together these results suggest that the JC measure effectively captures the degree of substitution between products. The results of the paper highlight a use-case of the JC measure in improving the predictive accuracy of a probabilistic demand model. However, together with the descriptive evidence comparing D_{Observed} to D_{JC} the paper shows that the JC measure can be used more generally as an effective measure of substitution when detailed product characteristics data are not available. The measure can be used in a variety of applications, such as in the construction of recommender systems and in the design of marketing strategies. By identifying products that are close substitutes, firms can target marketing campaigns to consumers who

have browsed these products. In addition, the JC measure can be used to construct recommender systems. By identifying products that are close substitutes, firms can recommend, for example through advertisements or the order of the user's search results, products that are likely to be purchased given the consumer's history of product browsing. This can increase the likelihood of a purchase and improve the user experience by enabling users to more quickly reach products they are interested in.

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Appendix
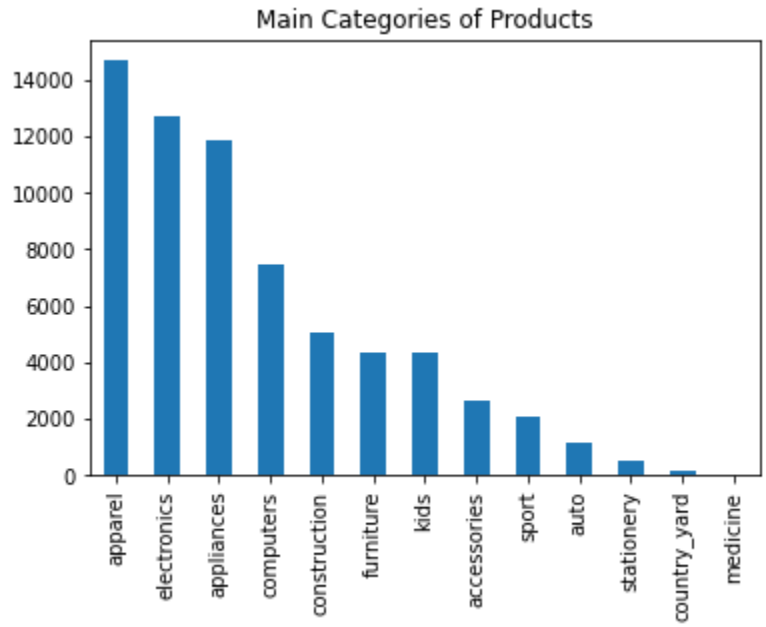
Figure 6: Main Categories of Products